

Spivak Chapter 3 Problems

December 30, 2025

1 Problem 1

Let $f(x) = 1/(1+x)$. What is

1. $f(f(x))$.

$$\frac{1}{1 + \frac{1}{1+x}}.$$

2. $f(\frac{1}{x})$.

$$\frac{1}{1 + \frac{1}{x}}.$$

3. $f(cx)$.

$$\frac{1}{1 + cx}.$$

4. $f(x+y)$.

$$\frac{1}{1 + x + y}.$$

5. $f(x) + f(y)$.

$$\frac{1}{1+x} + \frac{1}{1+y}.$$

6. For which numbers c is there a number x such that $f(cx) = f(x)$?

For all real c , if $x = 0$, $f(cx) = f(x)$.

7. For which numbers c is it true that $f(cx) = f(x)$ for two different numbers x ?

$c = 1$.

2 Problem 2

Let $g(x) = x^2$, and let

$$h(x) = \begin{cases} 0, & \text{if } x \text{ is rational,} \\ 1, & \text{if } x \text{ is irrational.} \end{cases}$$

1. For which y is $h(y) \leq y$?
 $h(y) \leq y$ for $y \notin \mathbb{Q}, y \geq 1$, and for $y \in \mathbb{Q}, y \geq 0$.
2. For which y is $h(y) \leq g(y)$?
 $h(y) \leq g(y)$ for all $y \in \mathbb{Q}$ and for all $y \notin \mathbb{Q}, |y| \geq 1$.
3. What is $g(h(z)) - h(z)$?
0.
4. For which w is $g(w) \leq w$?
 $g(w) \leq w$ for all $0 \leq w \leq 1$.
5. For which ϵ is $g(g(\epsilon)) = g(\epsilon)$?
 $\epsilon = -1, 0, 1$.

3 Problem 3

Find the domain of the functions defined by the following formulas.

1. $f(x) = \sqrt{1 - x^2}$.
 $x \in [-1, 1]$.
2. $f(x) = \sqrt{1 - \sqrt{1 - x^2}}$.
 $x \in [-1, 1]$.
3. $f(x) = \frac{1}{x-1} + \frac{1}{x-2}$.
 $x \neq 1, 2$.
4. $f(x) = \sqrt{1 - x^2} + \sqrt{x^2 - 1}$.
 $x = -1, 1$.
5. $f(x) = \sqrt{1 - x} + \sqrt{x - 2}$.
 $x \in \emptyset$.

4 Problem 4

Let $S(x) = x^2$, $P(x) = 2^x$, and let $s(x) = \sin x$. Find each of the following.

1. $(S \circ P)(y)$.
 2^{2y} .
2. $(S \circ s)(y)$.
 $\sin^2(y)$.
3. $(S \circ P \circ s)(t) + (s \circ P)(t)$.
 $2^{2 \sin t} + \sin 2^t$.
4. $s(t^3)$.
 $\sin t^3$.

5 Problem 8

For which numbers a, b, c , and d will the function

$$f(x) = \frac{ax + b}{cx + d}$$

satisfy $f(f(x)) = x$ for all x ?

$$x = f(f(x)) = \frac{a \frac{ax+b}{cx+d} + b}{c \frac{ax+b}{cx+d} + d},$$

$$x^2(ac + dc) + x(d^2 - a^2) + b(a - d) = 0.$$

Thus, either $a = d$ or $a = -d$. Then, $a = d = 0$, $a = d \neq 0$ and $b, c = 0$, or $a + d = 0$ and $f(x) = \frac{ax+b}{cx-a}$ for all $x \neq \frac{a}{c}$.

6 Problem 9

If A is any set of real numbers, define a function C_A as follows:

$$C_A(x) = \begin{cases} 1, & \text{if } x \text{ in } A, \\ 0, & \text{if } x \text{ not in } A. \end{cases}$$

Find expressions for $C_{A \cap B}$ and $C_{A \cup B}$ and C_{R-A} , in terms of C_A and C_B .

$$C_{A \cap B} = \begin{cases} 1, & \text{if } x \text{ in } A \text{ and } x \text{ in } B, \\ 0, & \text{if } x \text{ not in } A \text{ or } x \text{ not in } B. \end{cases}$$

Expressed algebraically, $C_A C_B$.

$$C_{A \cup B} = \begin{cases} 1, & \text{if } x \text{ in } A \text{ or } x \text{ in } B, \\ 0, & \text{if } x \text{ not in } A \text{ and } x \text{ not in } B. \end{cases}$$

Expressed algebraically, $C_A + C_B - C_A C_B$.

$$C_{R-A} = \begin{cases} 1, & \text{if } x \text{ not in } A, \\ 0, & \text{if } x \text{ in } A. \end{cases}$$

Expressed algebraically, $1 - C_A$.

7 Problem 12

A function f is even if $f(x) = f(-x)$ and odd if $f(x) = -f(-x)$.

1. Determine whether $f + g$ is even, odd, or not necessarily either, in the four cases obtained by choosing f even or odd, and g even or odd.

If f is even and g is even, $f + g$ is even, as $(f + g)(-x) = f(-x) + g(-x) = f(x) + g(x) = (f + g)(x)$. Likewise, if f is odd and g is odd, then $f + g$ is odd. If f is even and g is odd, then $f + g$ is not necessarily even or odd. If f is odd and g is even, then $f + g$ is not necessarily even or odd.

2. Do the same for $f \cdot g$.

If f is even and g is even, then $f \cdot g$ is even. If f is odd and g is odd, then $f \cdot g$ is even. If f is even and g is odd, then $(f \cdot g)(-x) = f(-x) \cdot g(-x) = f(x) \cdot (-g(x)) = -(f \cdot g)(x)$, so $f \cdot g$ is odd. Likewise, $g \cdot f$ is odd.

3. Do the same for $f \circ g$.

If f is even and g is even, then $f \circ g$ is even. If f is odd and g is odd, then $f \circ g$ is odd. If f is odd and g is even, then $f \circ g$ is even. If f is even and g is odd, then $f \circ g$ is even.